DMRG in era of MPS.

- → what is MPS?
- → what is DMRG?
- → how do they come together?

for N-body wavefunction, dim(Hilbert space $\mathcal{H}$) = d^N (spin → 2^N)
↳ not computationally efficient!

Assumption: low energy, low entanglement states of Hilbert space can be represented as a MPS.

$$|\psi\rangle = \sum^{\sigma} A^{i} \cdots A^{n} |i \cdots n\rangle$$

Why low entanglement? GS of local hamiltonian is less entangled than random state. → AREA LAW

Local hamiltonian: $S_A \propto L^{d-1}$
Random: $S_A \propto L^{d}$
} surface vs volume scaling → only boundary of

Notations

c → scalar $\qquad$ A_{ij} → matrix $\qquad$ each branch is another index.

v_i → vector $\qquad$ A_{ij}^{s} → tensors ...

Example contraction calculations:

$$\sum_{j,k} A_{ijk} B_{jklm} = T_{ilm} \implies$$

summed over / contracted

Complexity:

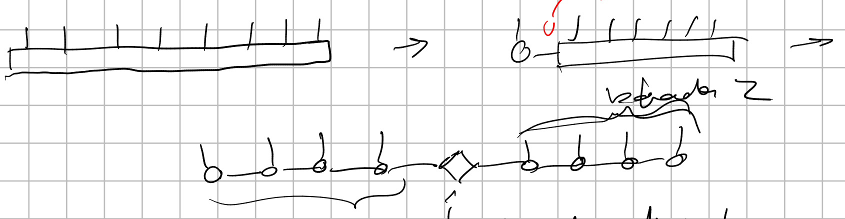
$$\prod_{\text{cont.}} D_i \cdot \prod_{\text{open}} D_j \Rightarrow$$ product of dimensionality of each "leg"

How do we get MPS from wavefunction? SVD!

→ Using SVD, keep decomposing until we get Schmidt decomposition.

$$|\psi\rangle = \sum \psi_{ij} \underbrace{|\alpha\rangle_i |\beta_j\rangle}_{\text{bipartite.}}$$ SVD constantly, from left → right

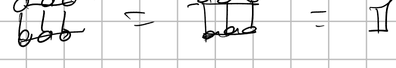
SVD from left $\qquad$ SVD from right



subsystem 2

subsystem 1 $\qquad$ singular value!

where

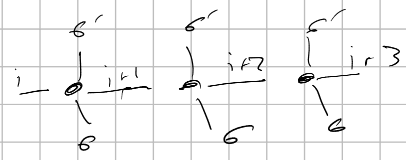


$= \mathbb{I}$

If wavefunctions become MPS, Hamiltonian become MPOs

MPO:

$$W^{\sigma_1,\sigma_1'}_{i,i+1} \; W^{\sigma_2,\sigma_2'}_{i+1,i+2} \; W^{\sigma_3,\sigma_3'}_{i+2,i+3} \; \dots \Rightarrow$$



We need to define behaviour for 3 cases:

We do this in the following manner: consider all possible actions of Hamiltonian:

Paths on one site:

e.g. $H = J\,\sigma_i^+ \sigma_{i+1}^- + h\,\sigma_i^z$

	Block A	SU	Block B	
	— I —	σ^z	— I —	
1	— I —	σ^+	σ^+ —	1
2	— σ^+	σ^+	—	2
3	— x —	—	—	3
	—	—	— x —	

all possible terms for each; to sum through.

these are our bases.

3 diff. unique entries for each: 3×3

$$\begin{array}{c|c} & \text{SU} \\ \begin{matrix} \text{L} \\ -x- \\ -J\sigma^+ \\ -I- \end{matrix} & \begin{bmatrix} I & 0 & 0 \\ \sigma^+ & 0 & 0 \\ \sigma^z & J\sigma^+ & I \end{bmatrix} \\ & \begin{matrix} -I- & \sigma^x- & -x- \end{matrix} \end{array}$$

MPO!

(R

More exercises:

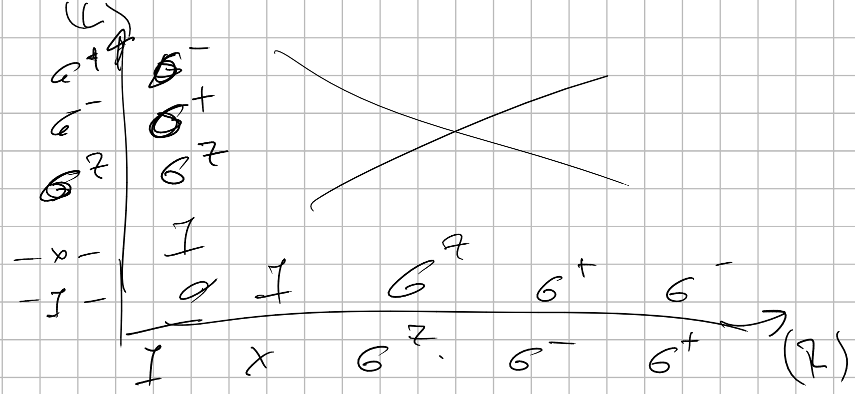
$S_1 - S_2$:

$$H_{S_{1,2}} = J\,\frac{1}{2}\left(\sigma_j^+ \sigma_{j+1}^- + \sigma_j^- \sigma_{j+1}^+\right) + \sigma_j^z \sigma_{j+1}^z$$

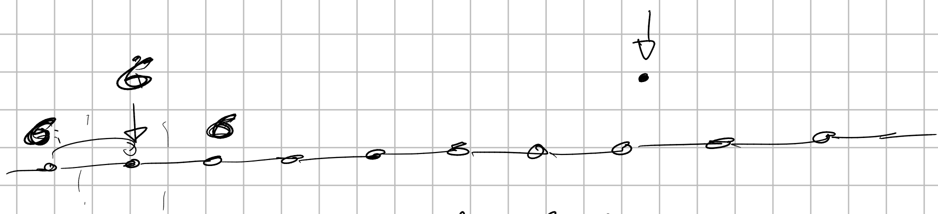
— I —	$\frac{1}{2}J\sigma_j^+$	σ_{j+1}^- —
— $\frac{1}{2}J\sigma_j^+$	σ_{j+1}^-	— I —
— I —	$\frac{1}{2}J\sigma_j^-$	σ_{j+1}^+ —
— $\frac{1}{2}J\sigma_j^-$	σ_{j+1}^+	— I —

— I —	$J\sigma_j^z$	σ_{j+1}^z —
— $J\sigma_j^z$	σ_{j+1}^z	— I —
— I —	I	— x —
— x —	I	— I —

Simplify:

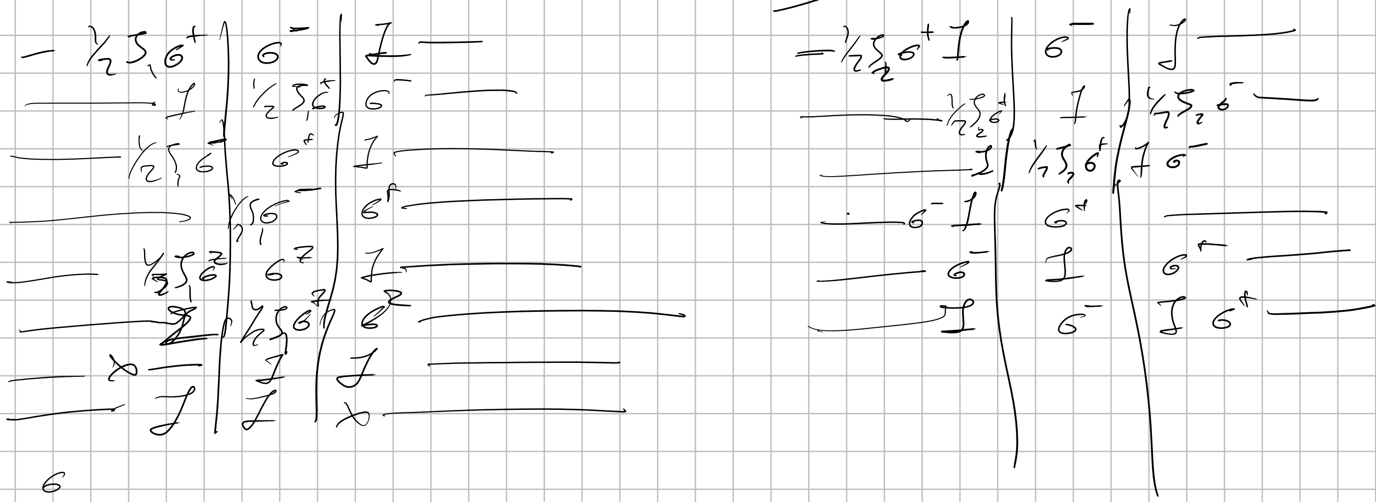


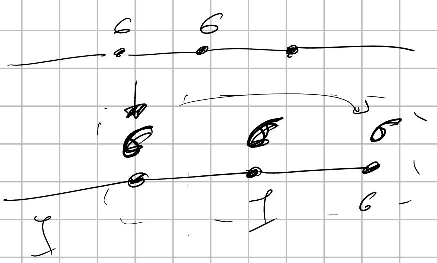
$$H = \zeta_1 \, \sigma_i \cdot \sigma_{i+1} + \zeta_2 \, \sigma_i \cdot \sigma_{i+2}$$

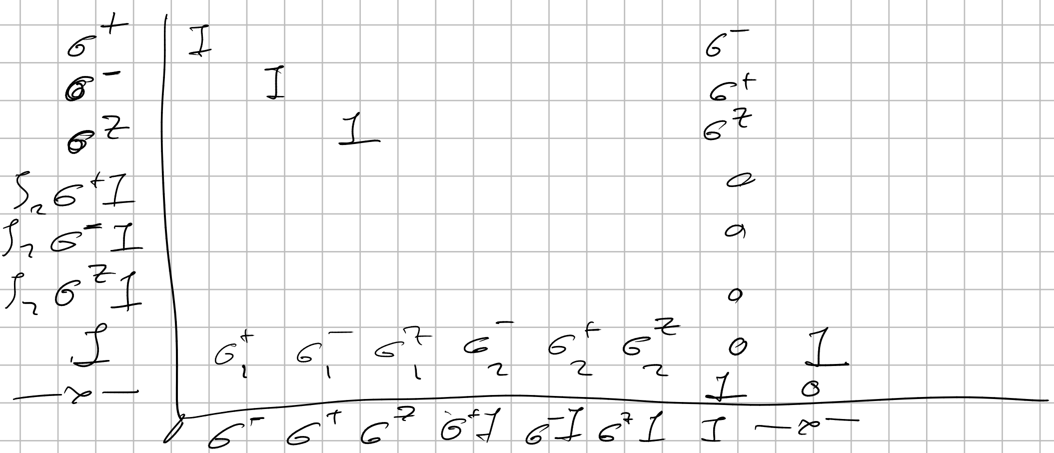


Add ζ_1:

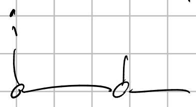
then ζ_2:



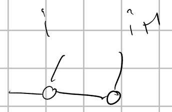




What about boundary case?



$$W_1 = \begin{matrix} \sigma^- \\ \sigma^+ \\ \sigma^z \end{matrix}$$



$$J\sum(\sigma_i \cdot \sigma_{i+1}) = J\tfrac{1}{2}(\sigma^+\sigma^- + \sigma^-\sigma^+) + J\sigma^z\sigma^z$$

i	$i+1$
σ^-	σ^+
σ^+	σ^-
σ^z	σ^z
0	I
I	$-r$

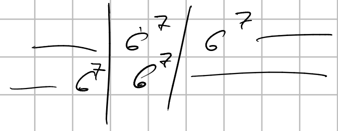
$$W_1 = \begin{pmatrix} \sigma^- \\ \sigma^+ \\ \sigma^z \\ I \\ 0 \end{pmatrix}$$

$$W_L = \begin{pmatrix} \sigma^+ \\ \sigma^- \\ \sigma^z \\ 0 \\ I \end{pmatrix}$$

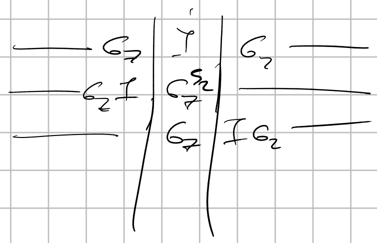
From paper:

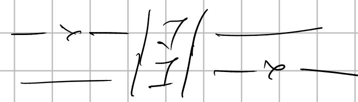
$$\hat{H} = J_1 \sum S_i^z S_{i+1}^z + J_2 \sum S_i^z S_{i+2}^z$$

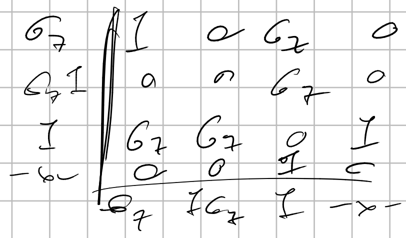
J_1:



J_2:



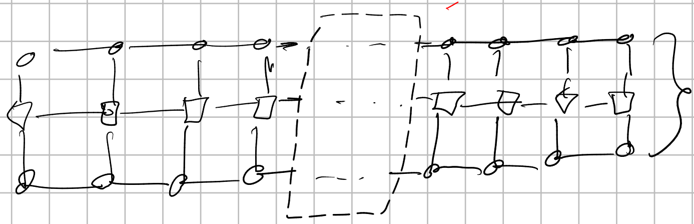




DMRG

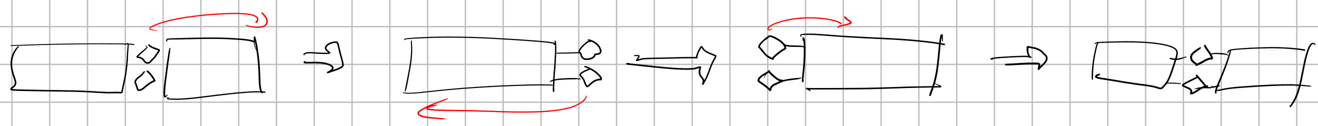
Take our constructed MPO and:

SVD



diagonalize, SVD, optimize

We then sweep this, going as:



Why sweep twice? normalize from left + right!

Calculating observables = recall that:

These nodes / matrices are simply a transformation of the basis. Contain 0 information about |ψ⟩ whatsoever

Singular values / Schmidt values tell us everything about the system:
→ ground state, few are.

Computing observables is quite easy: